

Multimodal Integration of EEG and Contextual Data for Enhanced Epilepsy Classification and Prediction

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Abstract

Epilepsy affects over 50 million people worldwide, yet accurate diagnosis remains challenging when EEG appears visually normal during interictal periods.

Research Gap:

- Traditional EEG-only approaches miss subtle biomarkers
- Contextual factors (sleep, stress, activity) provide complementary evidence
- Need for unified framework integrating both data streams

Proposed Solution:

Integrate interictal EEG with smartphone-derived contextual data using multimodal machine learning to improve epilepsy classification and seizure prediction.



Problem Statement

Clinical Challenge

EEG-only models struggle to detect subtle, hidden patterns in interictal recordings, while trigger-only data lacks direct neurophysiological evidence.

EEG Limitations

- Visually normal during interictal periods
- Misses subtle biomarkers
- Limited temporal coverage

Trigger Data Potential

- Continuous monitoring capability
- Captures lifestyle factors
- Environmental influences

Research Question:

"Can multimodal integration of interictal EEG and smartphone-derived seizure trigger data improve epilepsy classification and provide reliable seizure prediction compared to EEG-only approaches?"



Literature Review

- Rudroff (2025): AI-based digital biomarkers for fatigue monitoring.
- Abadal et al. (2024): Graph neural networks for electroencephalogram analysis: Alzheimer's disease and epilepsy use cases.
- Xiao et al. (2023): Self-supervised Learning with Attention Mechanism for EEG-based seizure detection.

Stubberud, A., Ingvaldsen, S. H., Brenner, E., Winnberg, I., Olsen, A., Gravdahl, G. B., Matharu, M. S., Nachev, P., & Tronvik, E. (2023). Forecasting migraine with machine learning based on mobile phone diary and wearable data. *Cephalalgia*, 43(5), 1–10. <https://doi.org/10.1177/03331024231169244>



Research Objectives

1. Multimodal Framework Development

Develop a multimodal deep learning framework for integrating EEG and trigger data using attention-based fusion mechanisms.

2. Interictal Biomarker Extraction

Extract subtle interictal biomarkers using deep and hybrid models (CNNs, Transformers, GNNs).

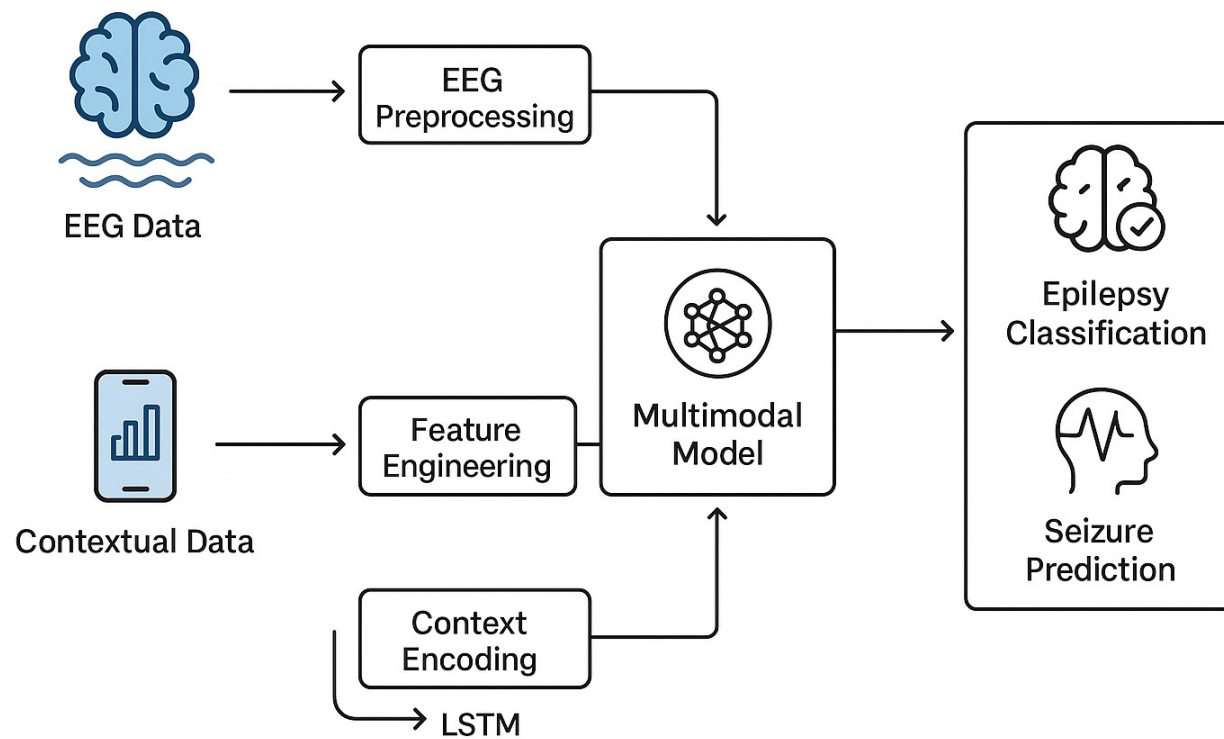
3. Trigger Integration

Incorporate lifestyle/environmental triggers (sleep, stress, activity) into predictive modeling using LSTMs/Transformers.

4. Comparative Evaluation

Compare multimodal models with unimodal baselines using cross-dataset validation and ablation studies.

Multimodal Integration of EEG and Contextual Data





Methodology - Data Sources

EEG Data

Datasets:

- TUH EEG Corpus
- CHB-MIT Database
- EPILEPSIAE

Types: Interictal segments from epilepsy patients + healthy controls

Preprocessing: Artifact removal, band-pass filtering, ICA

Contextual Data

Sources:

- Smartphone sensors
- Wearable devices
- Patient diaries
- Environmental APIs

Measures: Sleep quality, HRV-based stress, physical activity, environmental conditions

Note: Initial development uses simulated/secondary source data. Future work will collect real contextual data under IRB approval with informed consent.



Methodology - EEG Processing Pipeline

Step 1: Preprocessing

Filtering

Band-pass 0.5-32 Hz

Artifact Removal

ICA decomposition

Normalization

Z-score standardization

Step 2: Feature Extraction

Traditional Features:

- Time-domain: Mean, variance, skewness
- Frequency-domain: PSD bands
- Entropy: Approximate, sample entropy
- Connectivity: PLI, coherence

Deep Learning:

- CNNs: Automated feature learning
- Transformers: Attention mechanisms
- Latent embeddings: 256-dimensional



Methodology - Trigger Data Processing

Smartphone Data Collection

Passive Monitoring:

- Accelerometer (activity patterns)
- GPS (mobility tracking)
- Screen usage (sleep inference)
- Device interactions

Active Assessment:

- Ecological momentary assessments
- Patient diary entries
- Cognitive tests
- Voice recordings

Feature Engineering

Sleep

Duration, efficiency, fragmentation

Stress

HRV metrics from PPG

Activity

Step count, motion intensity

Environment

Weather, pollution APIs

Temporal Modeling:

LSTM/Transformer layers to capture time dependencies and cumulative effects (e.g., sleep deficit over days)



Methodology - Multimodal Fusion

Early Fusion

Concatenate EEG embeddings and trigger embeddings into joint feature space before classification.

Pros: Learns interactions early | Cons: High dimensionality

Late Fusion

Train separate models for EEG and triggers, combine outputs using ensemble methods (voting, averaging).

Pros: Modality independence | Cons: Limited interaction learning

Hybrid Fusion with Attention

Adaptive weighting mechanism assigns importance to EEG vs. trigger signals based on context.

```
 $\alpha_{\text{EEG}}, \alpha_{\text{trigger}} = \text{SoftMax}(W \cdot [h_{\text{EEG}}; h_{\text{trigger}}])$   
 $h_{\text{fused}} = \alpha_{\text{EEG}} \cdot h_{\text{EEG}} + \alpha_{\text{trigger}} \cdot h_{\text{trigger}}$ 
```

Proposed: Hybrid approach with cross-modal attention for optimal performance



Methodology - Classification & Prediction

Classification Task

Distinguish interictal epileptic EEGs from healthy control EEGs

Input: Fused EEG + Trigger features

Output: Binary/Multi-class labels

Loss: Cross-entropy

Prediction Task

Estimate seizure likelihood by aligning trigger history with interictal EEG

Input: Temporal sequences

Output: Risk probability [0,1]

Loss: MSE/Binary cross-entropy

Evaluation Metrics

Accuracy
Overall correctness

F1-Score
Precision-Recall balance

AUROC
Discrimination ability

Sensitivity
True positive rate

Specificity
True negative rate



Validation Strategy

Cross-Validation

5-fold or 10-fold cross-validation within each dataset to assess model stability and prevent overfitting.

Cross-Dataset Testing

Train on one dataset (e.g., CHB-MIT), test on another (e.g., TUH) to evaluate generalizability across different acquisition protocols.

Ablation Studies

- EEG-only model (baseline)
- Trigger-only model (baseline)
- Multimodal model (proposed)

Measure contribution of each modality and fusion strategy



Expected Contributions

1 **Novel Multimodal Framework**

First comprehensive integration of EEG biomarkers with smartphone-derived contextual triggers using advanced fusion techniques.

2 **Hidden Biomarker Identification**

Discovery of interictal EEG patterns contextualized by lifestyle and environmental conditions that are invisible to conventional analysis.

3 **Digital Phenotyping Validation**

Demonstrate how smartphone-based digital phenotyping complements traditional neurophysiological assessment.

4 **Clinical Translation Path**

Proof-of-concept for practical smartphone-based epilepsy monitoring and prediction system for real-world deployment.



Limitations & Future Work

- Initial use of simulated trigger data.
- Limited dataset diversity (public EEG datasets).
- Potential computational complexity for real-time use.
- Future: Collect real data under IRB approval, extend to temporal forecasting, deploy mobile app.



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Thank You

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- Questions and Discussion



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